NATIONAL INSTITUTE OF TECHNOLOGY ANDHRA PARDESH
Department of Electrical Engineering

EE2621 –Introduction to Machine Learning

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Project :Human-Centric Chess AI Project

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1. Introduction

Chess is one of the most complex strategy games played by humans. Traditional chess engines focus only on winning by calculating millions of possible moves. However, these systems do not consider how humans think while playing chess. Human-Centric Chess AI is designed to understand and interact with players in a way that is closer to human thinking. Instead of only focusing on the strongest move, the system tries to explain moves, adapt to player skill level, and help players learn chess effectively.

2.The Problem Statement (The "Why")

Engines vs. Humans: Traditional engines (Stockfish) find the perfect move, which is often incomprehensible to humans.

The Gap: We lack tools that understand why a human makes a mistake or what a 1200-rated player is likely to do next.

Objective: To build a system that predicts human-like moves and quantifies player skill based on behavioral patterns.

Flow Chart :

Chess Board → TNN(Transformer Neural Network) → Human Move Prediction

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Compare with Stockfish

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Features (ACPL, move match rate)

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Random Forest → ELO Prediction

3.Key Libraries:

1.pip install python-chess (The backbone for rules and board state).

2.pip install torch torchvision.

3.pip install stockfish (for Comparing the human move and model prediction against the engine's "ground truth".)

4.pip install scikit-learn pandas numpy (For data handling and ELO regression).

4.UI Layout Sketch (In streamlit)

Left Column: The Board	Middle Column: Human Metrics	Right Column: ML Insights
Interactive Chessboard	**Current Move Eval (Stockfish)**	**Model Prediction**
Displays the current FEN state.	Shows the Centipawn Loss (CPL) for the last move.	"The 1500-ELO Model would play: **Nf3** with 82% confidence".
Move History List	**Error Classification**	**ELO Tracker**
Highlights selected moves for analysis.	Categorizes moves as "Inaccuracy," "Mistake," or "Blunder" based on CPL.	Dynamic update of estimated ELO using the Random Forest Regressor.

5.Project Architecture

High-Level Workflow:

1.Data: Lichess PGNs -> Tensor Representation.

2.Move Prediction: CNN/ResNet (Policy Network).

3.Analysis: Stockfish Evaluation ->Centipawn Loss.

4.Rating: Random Forest / Regression for ELO estimation.

6.Data Engineering

1 .Source: Lichess Open Database (segmented by ELO).

2.Preprocessing: Converting PGN to 8 X 8 X 12 bitmask tensors.

3.Feature Planes: Mention additional planes for castling rights, side to move, and en passant squares.

4.Cleaning: Filtering out ultra-short games or bullet games where "pre-moves" skew data.

7.Move Prediction Model

1.Goal: The "Maia" approach—mimicking human play at specific levels.

2.Architecture: **Transformer** Neural Network (CNN) / ResNet.

3.Output Layer: Softmax over 4096 possible from-to square combinations.

4.The Mask: Briefly mention the Legal Move Mask to ensure the model only suggests valid chess moves.

8.Game Analysis & Quality

1. Engine Integration: Using Stockfish API to get "Ground Truth" evaluations.
2. Metric: Centipawn Loss (CPL) and Average CPL (ACPL).
3. Classification Logic:

Inaccuracy: 50–100 CPL.

Mistake: 100–300 CPL.

Blunder: >300 CPL.

9.ELO Rating Estimation

1.Feature Engineering: Beyond just ACPL.

Consistency: How often the player finds the top 3 engine moves.

Time Management: Variability in move times (if available in dataset).

Opening Breadth: Variety of openings played.

Model: We are using Random Forest/Regression (better for tabular feature data).

10.Preliminary Results / Expected Output

Move Accuracy: "The model predicts the exact human move X% of the time."

Skill Sensitivity: "The 1100-ELO model predicts 1100-rated moves better than it predicts 2000-rated moves" (The Maia benchmark).

Visualization: (Insert a screenshot of a chessboard showing predicted vs. actual moves).

11. Technical Challenges & Solutions

Computation: Handling millions of chess positions (Solution: HDF5/NPZ pre-processing).

Evaluation Latency: Stockfish is slow (Solution: Multi-threading or pre-analyzed datasets).

Class Imbalance: High-rated players blunder less; handling the scarcity of "blunder data."

Key References

McIlroy-Young, R. (2020). Aligning Superhuman AI with Human Behavior (Maia Chess).

Silver, D. (2017). Mastering Chess by Self-Play (AlphaZero).

Lichess.org Open Database (2025).

12.Conclusion & Future Scope

Summary: We’ve moved from "Chess as a Math Problem" to "Chess as a Behavioral Study."

Alignment: Contributes to the broader ML goal of aligning superhuman AI with human behavior, making AI more relatable and useful in collaborative environments.